Low Level Design (LLD) – Azure Multi-Region 3-Tier Application Architecture

1. Overview

This document describes the Low Level Design (LLD) for a highly available and secure 3-tier application deployed on Microsoft Azure across two regions:

- India Region (Primary)
- USA Region (Secondary / DR)

The architecture supports:

- Secure connectivity from On-Premises to Azure
- Multi-region deployment
- Network isolation using subnets and NSGs
- Application delivery using Azure Application Gateway
- Frontend, Backend, and Database tier separation
- Secrets management using Azure Key Vault
- Infrastructure provisioning using Terraform and Azure Storage Account backend

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2. High Level Components

External Components

Component	Purpose
On-Prem Datacenter	Enterprise internal network
VPN Gateway	Secure tunnel between On-Prem and Azure
Front Door	Global traffic routing and failover
Users	End users accessing the application

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3. Regional Architecture

Both India and USA regions contain:

- Separate Virtual Network (VNet)
- Azure Firewall
- Frontend Subnet
- Backend Subnet
- Database Subnet
- Bastion Subnet
- Internal Load Balancer
- Application Gateway
- Key Vault
- Bastion Host
- Frontend VM
- Backend VM
- Database VM

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4. Network Design

4.1 India Region Network

VNet

Parameter	Value
VNet Name	india-prod-vnet
Address Space	10.10.0.0/16
Region	Central India

Subnets

Subnet	CIDR	Purpose
frontend-subnet	10.10.1.0/24	Frontend VMs and App Gateway
backend-subnet	10.10.2.0/24	Backend APIs and services
database-subnet	10.10.3.0/24	Database VMs
AzureBastionSubnet	10.10.4.0/27	Azure Bastion
GatewaySubnet	10.10.5.0/27	VPN Gateway
firewall-subnet	10.10.6.0/26	Azure Firewall

4.2 USA Region Network

VNet

Parameter	Value
VNet Name	usa-prod-vnet
Address Space	10.20.0.0/16
Region	East US

Subnets

Subnet	CIDR	Purpose
frontend-subnet	10.20.1.0/24	Frontend VMs and App Gateway
backend-subnet	10.20.2.0/24	Backend APIs and services
database-subnet	10.20.3.0/24	Database VMs
AzureBastionSubnet	10.20.4.0/27	Azure Bastion
GatewaySubnet	10.20.5.0/27	VPN Gateway
firewall-subnet	10.20.6.0/26	Azure Firewall

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5. Connectivity Design

5.1 On-Prem to Azure Connectivity

Connectivity Type

- Site-to-Site VPN

Flow

On-Prem → VPN Gateway → Azure Firewall → Application Components

VPN Configuration

Parameter	Value
VPN Type	Route Based
SKU	VpnGw2
Redundancy	Active-Active
BGP	Enabled

5.2 Global Access Flow

Users → Azure Front Door → Regional Application Gateway → Frontend VM → Backend VM → Database VM

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6. Traffic Flow

6.1 User Request Flow

1. User accesses public application URL.
2. Azure Front Door receives the request.
3. Front Door routes traffic to nearest healthy region.
4. Request reaches Azure Application Gateway.
5. Application Gateway performs WAF inspection.
6. Request forwarded to Frontend VM.
7. Frontend communicates with Backend VM through Internal Load Balancer.
8. Backend accesses Database VM.
9. Response returned back to user.

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7. Security Design

7.1 Azure Firewall

Purpose

- Centralized ingress and egress filtering
- Threat protection
- Network traffic inspection

Rules

Rule Type	Allowed Traffic
DNAT	HTTPS traffic to App Gateway
Network Rule	Frontend to Backend
Network Rule	Backend to Database
Application Rule	Outbound internet access

7.2 Network Security Groups (NSGs)

Frontend NSG

Source	Destination Port	Action
Internet	443	Allow
App Gateway	80/443	Allow
Any	Any	Deny

Backend NSG

Source	Destination Port	Action
Frontend Subnet	8080	Allow
Internal LB	8080	Allow
Any	Any	Deny

Database NSG

Source	Destination Port	Action
Backend Subnet	1433/3306	Allow
Any	Any	Deny

7.3 Azure Bastion

Purpose

- Secure VM access without public SSH/RDP exposure

Access Method

- Browser-based secure RDP/SSH

7.4 Azure Key Vault

Stored Secrets

- Database passwords
- API keys
- SSL certificates
- Service principal secrets

Access Control

- Managed Identity
- RBAC enabled

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8. Compute Layer

8.1 Frontend VM

Parameter	Value
VM Type	Standard_B2s
OS	Ubuntu/Linux
Purpose	Web/UI Hosting
Public IP	Enabled

Installed Components

- NGINX
- Frontend application
- Monitoring agent

8.2 Backend VM

Parameter	Value
VM Type	Standard_B2s
OS	Ubuntu/Linux
Purpose	API & Business Logic
Public IP	Disabled (Preferred)

Installed Components

- Java/.NET/Node runtime
- Backend APIs
- Monitoring agent

8.3 Database VM

Parameter	Value
VM Type	Standard_D4s_v5
OS	Linux/Windows
Purpose	Database Services
Public IP	Disabled

Database Options

- MySQL
- PostgreSQL
- SQL Server

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9. Load Balancing Design

9.1 Azure Front Door

Purpose

- Global routing
- Health probe monitoring
- Automatic failover
- SSL termination

Routing Policy

Scenario	Action
India Healthy	Route to India
India Down	Failover to USA
USA Healthy	Route to USA

9.2 Application Gateway

Features

- Layer 7 Load Balancing
- Web Application Firewall (WAF)
- SSL Offloading
- URL-based routing

WAF Mode

- Prevention Mode

9.3 Internal Load Balancer

Purpose

- Backend VM traffic distribution
- Internal-only communication

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10. Disaster Recovery Design

DR Strategy

- Active-Passive deployment
- India acts as Primary region
- USA acts as DR region

Failover Process

1. Front Door health probe detects outage.
2. Traffic automatically redirected to USA.
3. Backend services continue from DR region.

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11. Monitoring and Logging

Azure Monitor

Monitored Resources

- VMs
- VPN Gateway
- Firewall
- App Gateway
- Load Balancer

Log Analytics Workspace

Logs Collected

- Security logs
- VM metrics
- Application logs
- Firewall logs
- NSG flow logs

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12. Terraform Deployment Structure

Recommended Repository Structure

```
terraform/
│
├── modules/
│   ├── network/
│   ├── vm/
│   ├── firewall/
│   ├── application_gateway/
│   ├── bastion/
│   ├── keyvault/
│   └── monitoring/
│
├── env/
│   ├── india-prod/
│   └── usa-dr/
│
├── backend.tf
├── providers.tf
├── variables.tf
├── outputs.tf
└── main.tf
```

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13. Terraform Backend Configuration

Azure Storage Backend

```
terraform {
  backend "azurerm" {
    resource_group_name  = "tfstate-rg"
    storage_account_name = "prodtfstate"
    container_name       = "tfstate"
    key                  = "prod.terraform.tfstate"
  }
}
```

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14. Availability and Scalability

High Availability

- Multi-region deployment
- Front Door failover
- Multiple backend instances
- Redundant VPN Gateway

Scalability

- VM Scale Sets can be introduced
- Autoscaling supported
- App Gateway autoscaling enabled

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15. Best Practices

Security

- No direct SSH/RDP exposure
- Use Bastion access only
- Store secrets in Key Vault
- Enable NSG flow logs
- Enable WAF

Operations

- Use Infrastructure as Code
- Separate environments using workspaces
- Use remote backend state locking
- Enable resource tagging

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16. Recommended Azure Resource Naming Convention

Resource Type	Naming Pattern
Resource Group	rg--
VNet	vnet--
Subnet	snet--
VM	vm---01
NSG	nsg--
App Gateway	agw-
Firewall	afw-
Key Vault	kv--

Example:

- rg-prod-india
- vm-backend-india-01
- nsg-db-india

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17. Sample Azure Resource Configurations

17.1 Resource Group

```
resource "azurerm_resource_group" "india_rg" {
  name     = "rg-prod-india"
  location = "Central India"
}

resource "azurerm_resource_group" "usa_rg" {
  name     = "rg-prod-usa"
  location = "East US"
}
```

17.2 Virtual Network

```
resource "azurerm_virtual_network" "india_vnet" {
  name                = "india-prod-vnet"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name
  address_space       = ["10.10.0.0/16"]
}
```

17.3 Frontend Subnet

```
resource "azurerm_subnet" "frontend_subnet" {
  name                 = "frontend-subnet"
  resource_group_name  = azurerm_resource_group.india_rg.name
  virtual_network_name = azurerm_virtual_network.india_vnet.name
  address_prefixes     = ["10.10.1.0/24"]
}
```

17.4 Backend Subnet

```
resource "azurerm_subnet" "backend_subnet" {
  name                 = "backend-subnet"
  resource_group_name  = azurerm_resource_group.india_rg.name
  virtual_network_name = azurerm_virtual_network.india_vnet.name
  address_prefixes     = ["10.10.2.0/24"]
}
```

17.5 Database Subnet

```
resource "azurerm_subnet" "database_subnet" {
  name                 = "database-subnet"
  resource_group_name  = azurerm_resource_group.india_rg.name
  virtual_network_name = azurerm_virtual_network.india_vnet.name
  address_prefixes     = ["10.10.3.0/24"]
}
```

17.6 Network Security Group

```
resource "azurerm_network_security_group" "frontend_nsg" {
  name                = "nsg-frontend-india"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name

  security_rule {
    name                       = "Allow-HTTPS"
    priority                   = 100
    direction                  = "Inbound"
    access                     = "Allow"
    protocol                   = "Tcp"
    source_port_range          = "*"
    destination_port_range     = "443"
    source_address_prefix      = "*"
    destination_address_prefix = "*"
  }
}
```

17.7 Public IP

```
resource "azurerm_public_ip" "frontend_pip" {
  name                = "frontend-public-ip"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name
  allocation_method   = "Static"
  sku                 = "Standard"
}
```

17.8 Frontend VM

```
resource "azurerm_linux_virtual_machine" "frontend_vm" {
  name                = "vm-frontend-india-01"
  resource_group_name = azurerm_resource_group.india_rg.name
  location            = azurerm_resource_group.india_rg.location
  size                = "Standard_B2s"
  admin_username      = "azureuser"

  network_interface_ids = [
    azurerm_network_interface.frontend_nic.id
  ]

  admin_ssh_key {
    username   = "azureuser"
    public_key = file("~/.ssh/id_rsa.pub")
  }

  os_disk {
    caching              = "ReadWrite"
    storage_account_type = "Standard_LRS"
  }

  source_image_reference {
    publisher = "Canonical"
    offer     = "0001-com-ubuntu-server-jammy"
    sku       = "22_04-lts"
    version   = "latest"
  }
}
```

17.9 Backend VM

```
resource "azurerm_linux_virtual_machine" "backend_vm" {
  name                = "vm-backend-india-01"
  resource_group_name = azurerm_resource_group.india_rg.name
  location            = azurerm_resource_group.india_rg.location
  size                = "Standard_B2s"
  admin_username      = "azureuser"

  disable_password_authentication = true
}
```

17.10 Database VM

```
resource "azurerm_windows_virtual_machine" "database_vm" {
  name                = "vm-db-india-01"
  resource_group_name = azurerm_resource_group.india_rg.name
  location            = azurerm_resource_group.india_rg.location
  size                = "Standard_D4s_v5"
  admin_username      = "azureadmin"
  admin_password      = "P@ssw0rd1234!"
}
```

17.11 Internal Load Balancer

```
resource "azurerm_lb" "internal_lb" {
  name                = "backend-ilb"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name
  sku                 = "Standard"

  frontend_ip_configuration {
    name                          = "private-ip"
    subnet_id                     = azurerm_subnet.backend_subnet.id
    private_ip_address_allocation = "Dynamic"
  }
}
```

17.12 Azure Application Gateway

```
resource "azurerm_application_gateway" "appgw" {
  name                = "agw-india"
  resource_group_name = azurerm_resource_group.india_rg.name
  location            = azurerm_resource_group.india_rg.location

  sku {
    name     = "WAF_v2"
    tier     = "WAF_v2"
    capacity = 2
  }

  gateway_ip_configuration {
    name      = "gateway-ip-config"
    subnet_id = azurerm_subnet.frontend_subnet.id
  }
}
```

17.13 Azure Firewall

```
resource "azurerm_firewall" "firewall" {
  name                = "afw-india"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name

  sku_name = "AZFW_VNet"
  sku_tier = "Standard"
}
```

17.14 VPN Gateway

```
resource "azurerm_virtual_network_gateway" "vpn_gateway" {
  name                = "vpn-gateway-india"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name

  type     = "Vpn"
  vpn_type = "RouteBased"

  active_active = true
  enable_bgp    = true

  sku = "VpnGw2"
}
```

17.15 Azure Bastion

```
resource "azurerm_bastion_host" "bastion" {
  name                = "bastion-india"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name

  ip_configuration {
    name                 = "configuration"
    subnet_id            = azurerm_subnet.bastion_subnet.id
    public_ip_address_id = azurerm_public_ip.bastion_pip.id
  }
}
```

17.16 Azure Key Vault

```
resource "azurerm_key_vault" "kv" {
  name                       = "kv-prod-india"
  location                   = azurerm_resource_group.india_rg.location
  resource_group_name        = azurerm_resource_group.india_rg.name
  tenant_id                  = data.azurerm_client_config.current.tenant_id
  sku_name                   = "standard"
  purge_protection_enabled   = true
  soft_delete_retention_days = 7
}
```

17.17 Azure Front Door

```
resource "azurerm_cdn_frontdoor_profile" "frontdoor" {
  name                = "afd-global-prod"
  resource_group_name = azurerm_resource_group.india_rg.name
  sku_name            = "Standard_AzureFrontDoor"
}
```

17.18 Storage Account for Terraform Backend

```
resource "azurerm_storage_account" "tfstate" {
  name                     = "prodtfstate"
  resource_group_name      = azurerm_resource_group.india_rg.name
  location                 = azurerm_resource_group.india_rg.location
  account_tier             = "Standard"
  account_replication_type = "LRS"
}

resource "azurerm_storage_container" "tfstate_container" {
  name                  = "tfstate"
  storage_account_name  = azurerm_storage_account.tfstate.name
  container_access_type = "private"
}
```

17.19 Log Analytics Workspace

```
resource "azurerm_log_analytics_workspace" "law" {
  name                = "law-prod-india"
  location            = azurerm_resource_group.india_rg.location
  resource_group_name = azurerm_resource_group.india_rg.name
  sku                 = "PerGB2018"
  retention_in_days   = 30
}
```

17.20 Diagnostic Settings

```
resource "azurerm_monitor_diagnostic_setting" "firewall_logs" {
  name                       = "firewall-diagnostics"
  target_resource_id         = azurerm_firewall.firewall.id
  log_analytics_workspace_id = azurerm_log_analytics_workspace.law.id

  enabled_log {
    category = "AzureFirewallApplicationRule"
  }

  metric {
    category = "AllMetrics"
  }
}
```

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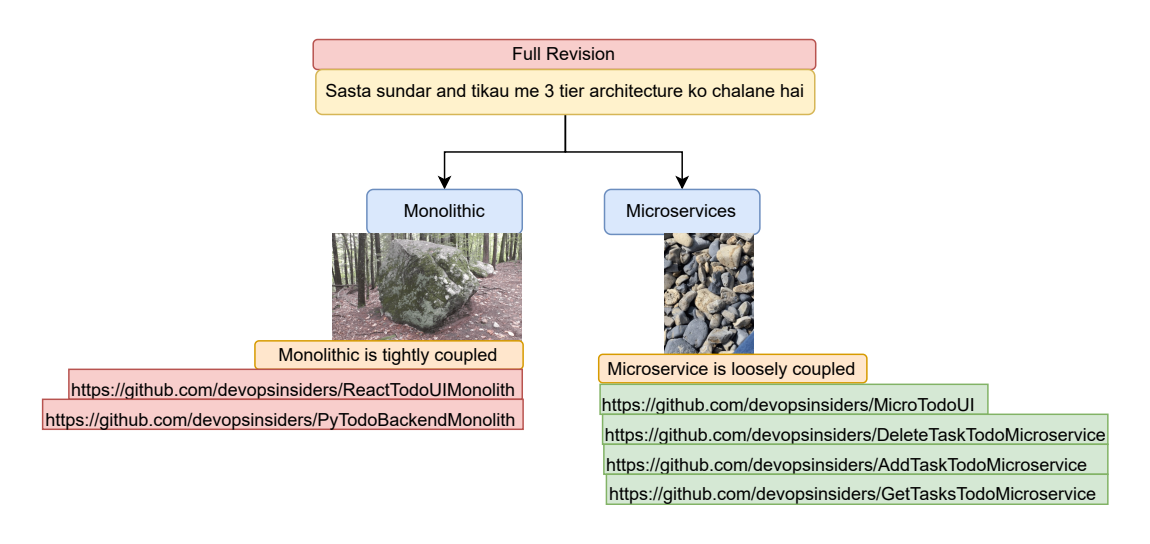
18. Conclusion

This architecture provides:

- High availability
- Disaster recovery capability
- Secure network isolation
- Controlled access management
- Centralized monitoring
- Scalable infrastructure
- Terraform-based Infrastructure as Code deployment

The design is suitable for enterprise-grade production workloads running on Azure with multi-region resiliency and secure hybrid connectivity.



Full Revision
Sasta sundar and tikau me 3 tier architecture ko chalane hai
Monolithic
Microservices
Monolithic is tightly coupled
https://github.com/devopsinsiders/ReactTodoUIMonolith
https://github.com/devopsinsiders/PyTodoBackendMonolith
Microservice is loosely coupled
https://github.com/devopsinsiders/MicroTodoUI
https://github.com/devopsinsiders/DeleteTaskTodoMicroservice
https://github.com/devopsinsiders/AddTaskTodoMicroservice
https://github.com/devopsinsiders/GetTasksTodoMicroservice

Discussion
1. SOW/BRD Documents
2. Functional Document
3. Assesment Phase/Requirement Gathering
3. HLD creation - architect decides cloud

Infra Design	App Design
LLD design	LLD design
Agile Planning	Agile Planning
Epic Creation (Big Feature)	Epic Creation (Big Feature)
User Stories me tod dete hai epic ko	User Storiesy me tod dete hai epic ko
Sprint Planning	Sprint Planning
IMplementation Phase	IMplementation Phase
User Story Broken into tasks	User Story Broken into tasks

BusinessRM team requirement gather karega

Implementation Phase for monolithic application
- Azure Cloud
- Terraform or Bicep
- Code rakhne ke lie Azure Repo ya Github (Trunk based brancing strategy)
- Pipeline ke lie Azure Pipeline ya Github Actions
- Security Tool -tfsec, trufflehog, superlinter, infracost, terratest, chefinspect

Provisioning of Landing Zone on Azure Cloud

Terraform Implementation

Jitne bhi resource block hai unka child module bn jata hai
- **1 Parent Module and Child Module creation**
- **Prerequisites set kia**
- **Terraform me coding block se block mila ke hua**
- **Foreach, map, dynamic block, conditional, optional attribute use karke generic module bana daale**
- **Child Modules ko versioning set karke repo me rakh dia**
- **Trunk based strategy se ser kar dia**
- **PR pipeline laga dia, linting, validation sab check hoga**
- **Fir dev me deployment hoga, then qa prod**

Pipeline
- **Pehle agent set kara**
- ADO ko azure se connection banane ke lie service connection banaenge
- **YAML pipeline likh ke sab steps usme dal dia**
- **stages, jobs and steps**
- **isse infra pipeline banega**
- Fir application pipeline banake infra pr code deploy kar denge

deployment strategy
- **Blue Green**
- **Canary**

Azure Kubernetes Services

Landing Zone
- Normal
- Hub and Spoke

Governance, Security, Networking, Identity, Compliances, Cost Management, Disaster Recovery
Ek bdia infra architect ka role hai secure infra banana
Scalable environment
Enterprise Governance

Azure ↔ **AWS** ↔ **GCP**

1. Identity Management
- Microsoft Entra ID
- Security policies for Conditional Access
- MFA
- PIM (Privileged Identity Management)
- RBAC (Role Based Access Control)
- App Registrations / Managed Identity

2. Networking resources
- Virtual network - Isolated network banata hai
- Subnet - Virtual network ka tukda
- NSG - Traffic Control (inbound/Outbound)
- Route Table
- Azure Firewall
- Azure bastion
- Agar do network dono cloud same region me hai toh pr hai toh Vnet Peering
- Agar do network dono cloud different region me hai toh pr hai toh Global Vnet Peering
- On Prem ko Azure Vnet se jodne ke lie Site to Site VPN
- Taar dalna ko apne ghr se azure data center tk toh express route
- VPN Gateway hybrid connection
- Application gateway - layer 7
- Loadbalancer - layer 4
- Frontdoor - global Loadbalancing - Layer 7
- Private Endpoint for Private connectivity for paas services
- Private DNS zones
- DDOS protection
- Traffic manager - Global Loadbalancing - Domain based routing

3. Compute Services
- Virtual machines
- VMSS
- App Services Plan and Web App
- Azure Function - Serverless
- Azure Container Instances
- Azure Logic Apps
- Azure Kubernetes Services

4. Storage Services
- Storage Account - Blob, file, Table Queues
- Datalake Gen 2

5. Database Services
- Azure SQL Database
- SQL managed Instances
- Cosmos DB - NoSQL
- PostgreSQL, Redis Cache Etc..

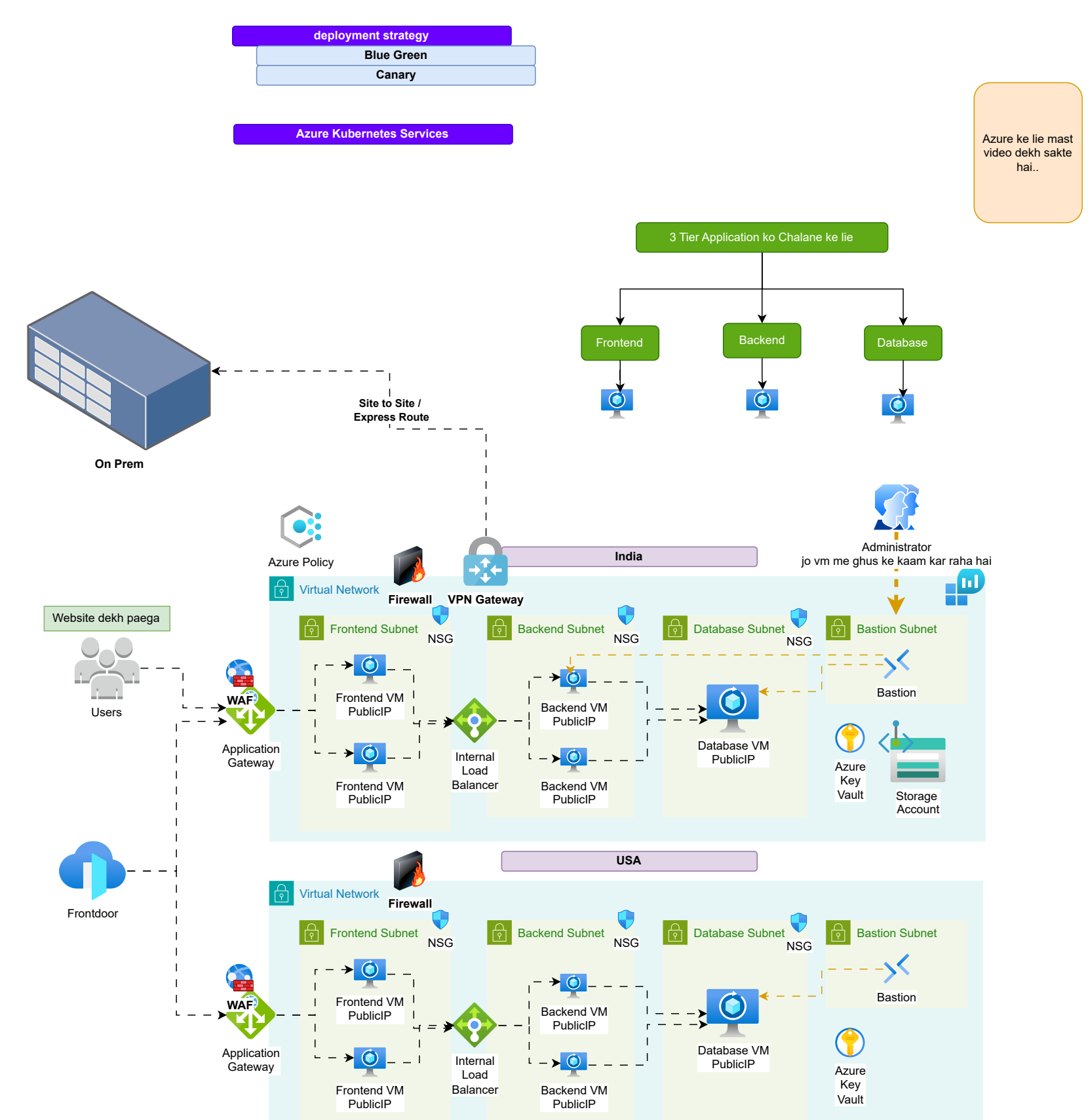
6. Security Services
- Defender
- Sentinel
- Key Vault, Azure policy
- Azure Blueprint
- Customer Managed Key

7. Monitoring Services
- Log Analytics Workspaces
- Azure Monitor
- Azure Application Insights
- Azure Alerts, Dashboards

Prometheus Grafana, datadog

8. Backup and DR
- Azure Site Recovery
- Recovery Service Vaults
- Geo Redundant Storage
- Backup Policy

Azure ke lie mast video dekh sakte hai.



Home Work

RG, Vnet, Subnet, VM, NSG, ASG, Firewall, Loadbalancer, Application Gateway, bastion Keyvault, Frontdoor, traffic manager, Sql Server, SQL DB, Storage accounts, Private Endpoint, Log Analytics Workspace, Azure Policy, Azure Backup, Azure Monitor

Document se, ChatGPT se, Youtube se kahi se bhi resource ke baare me padna hai...